HH-P-00430(GSA-FSS)
March 1, 1965
Revision of
Int. Fed. Spec. HH-R-00191b(GSA-FSS)
July 19, 1963 and
Interim Revision of
Fed. Spec. HH-R-191a
August 7, 1952.

INTERIM FEDERAL SPECIFICATION

PLASTIC MIX, REFRACTORY

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C. 20407 based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification HH-R-191a dated August 7, 1952.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic mix, refractory intended for use in industrial boiler furnaces and incinerators. In addition to covering high duty plastic refractories which are recommended for average service conditions, this specification also covers super duty plastic refractories. Super duty refractories are applied to advantage in the extreme temperature zones of furnaces where severe spalling and slagging occur. This specification includes the requirements of ASTM C 176-47.

1.2 Classification

1.2.1 Classes. Plastic mix, refractory covered by this specification shall be of the following classes as specified (see 6.2):

Class 1 - High duty
Class 2 - Super duty

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS.

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:
PPP-B-636 - Box, Fiberboard.
PPP-D-705 - Drum: Metal Shipping, Steel (over 12 and under 55-Gallon).
PPP-P-704 - Pails: Shipping, Steel (1 through 12 Gallon).

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

FSC 9350

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Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials (ASTM) Standards:
C 24-56 - Test for Pyrometric Cone Equivalent (PCE) of Refractory Materials.
C 92-46 - Test for Sieve Analysis and Water Content of Refractory Materials.
C 179-46 - Test for Combined Drying and Firing Shrinkage of Fireclay Plastic Refractories.
C 180-52 - Panel Spalling Test for Fireclay Plastic Refractories.
C 181-47 - Test for Workability Index of Fireclay Plastic Refractories.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 Material. The plastic refractory shall consist essentially of grog, raw clay, and water so proportioned as to result in a mixture of stiff mud consistency.

3.2 Water content. The plastic refractory, as received, shall contain not more than 15 percent of water calculated on the basis of wet weight (see 4.4.1).

3.3 Workability index. When tested as specified in 4.4.2, the workability index of the plastic refractory shall be not less than 15 percent nor more than 35 percent deformation. The test specimen shall stand 3 tamps of the test weight without crumbling.

3.4 Pyrometric cone equivalent (PCE).

3.4.1 Class 1. When tested as specified in 4.4.3, the pyrometric cone equivalent (PCE) of the class 1 plastic refractory shall be not lower than Number 31.

3.4.2 Class 2. When tested as specified in 4.4.3, the pyrometric cone equivalent (PCE) of the class 2 plastic refractory shall be not lower than Number 32-1/2.

3.5 Drying-and-firing shrinkage.

3.5.1 Class 1. The total drying-and-firing linear shrinkage of the class 1 plastic refractory shall be not more than 4.0 percent of the original length when tested after 1400° C. (2550° F) heating as specified in 4.4.4.

3.5.2 Class 2. The total drying-and-firing linear shrinkage of the class 2 plastic refractory shall be not more than 2.5 percent of the original length when tested after 1600° C. (2910° F) heating as specified in 4.4.4.

3.6 Panel spalling loss.

3.6.1 Class 1. The panel spalling loss of the class 1 plastic refractory shall be not more than 15.0 percent when tested after 1600° C. (2910° F) preheat in accordance with 4.4.5.

3.6.2 Class 2. The panel spalling loss of the class 2 plastic refractory shall be not more than 5.0 percent when tested after 1650° C. (3000° F) preheat in accordance with 4.4.5.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and test shall consist of all material of the same class submitted for delivery at one time.

4.3 Examination of the end item. Examination shall be made in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQL's) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled containers of refractory for examinations under 4.3.1, 4.3.2 and 4.3.3.

4.3.1 Examination of end item container for defects in appearance, workmanship, closures and marking. The sample unit for the examination specified in Table I shall be one filled container.

TABLE I. Examination of end item container for defects in appearance, workmanship, closures and marking.

Examine	Defect
Appearance and workmanship	Not type or size specified. Evidence of leaking.
Closures	End closures not as specified.
Markings	Omitted, illegible, incorrect, incomplete or not in accordance with contract requirements.

4.3.2 Examination of end item container for defects in appearance of refractory. The sample unit for the examination specified in Table II shall be one filled container.

TABLE II. Examination of end item container for defects in appearance of refractory.

Examine	Defect
Appearance	Material not as specified. Presence of foreign material Presence of large particles, lumps or caked material.

4.3.3 Examination of end item for defects in net contents. The sample unit for this examination shall be one filled container. The average net contents shall be not less than the specified or indicated quantity. Not less than five (5) sample units shall be drawn for this examination.

4.3.4 Inspection levels and acceptable quality levels (AQL's) for examinations. The inspection levels for determining the sample size and the acceptable quality levels (AQL's), expressed in defects per 100 units, shall be as in Table III.

TABLE III. Inspection levels and acceptable quality levels (AQL's) for examinations.

Examination paragraph	Inspection level	AQL
4.3.1	I	2.5
4.3.2	S-3	2.5
4.3.3	S-2	---

4.3.5 Testing of the end item. The end item shall be tested for the applicable characteristics as indicated in section 3 for each lot presented for examination. The sample unit shall be fifteen (15) pounds of plastic refractory. The inspection level for determining the sample size shall be S-1, except that no less than five (5) sample units shall be randomly selected throughout

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the lot. No more than one sample shall be drawn from any one container. The test sample shall consist of a seventy-five (75) pound composite of the sample units. Two or more determinations, as applicable, shall be performed on the composite. There shall be no evidence of failure to meet the specified requirements. The lot size shall be expressed in units of containers of refractory.

4.4 Test procedures.

4.4.1 Water content. Determine the water content in accordance with ASTM Method C 92-46.

4.4.2 Workability index. Determine the workability index of the plastic refractory in accordance with ASTM Method C 181-47.

4.4.3 Pyrometric cone equivalent. Determine the pyrometric cone equivalent of the plastic refractory in accordance with ASTM Method C 24-56.

4.4.4 Drying-and-firing shrinkage. Determine the drying-and-firing shrinkage of the plastic refractory in accordance with ASTM Method C 179-46.

4.4.5 Panel spalling loss. Determine the panel spalling loss in accordance with ASTM Method C 180-52.

5. PREPARATION FOR DELIVERY.

(For civil agency procurement, the definitions and application of the levels for packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Levels A and B.

5.1.1.1 One hundred-pound fiberboard box. One hundred pounds of the plastic refractory shall be packed in fiberboard boxes conforming to PPP-B-636. Boxes shall be provided with suitable vapor barriers.

5.1.1.2 One hundred-pound metal container. One hundred pounds of the plastic refractory shall be packed in metal pails conforming to PPP-P-704. Pails shall be provided with full removable heads.

5.1.1.3 Two hundred-pound metal container. Two hundred pounds of the plastic refractory shall be packed in metal drums conforming to PPP-D-705 or PPP-P-704. Drums shall be provided with full removable heads.

5.1.2 Level C. The plastic refractory shall be packed in a manner which will insure arrival at destination in a satisfactory condition and be acceptable to the carrier at the lowest rate.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES.

6.1 Intended use. The plastic refractories covered by this specification are used for the fabrication of furnace shapes and for the installation of specially prefabricated refractories. These materials are used to form monolithic furnace walls, arches, and boiler settings. At the furnace operating temperatures, the material facing the hot zone, vitrifies sufficiently to create a condition in which the refractory material approaches the strength of fireclay brick of corresponding quality.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Class of material required (see 1.2.1).

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(c) Level of packing and size of container required (see section 5).

6.3 Transportation description. Transportation descriptions and minimum weights applicable to these items are:

Rail:
Plastic fire brick
Carload minimum weight 90,000 pounds.

Motor:
Plastic fire brick
Truckload minimum weight 40,000 pounds, subject to Rule 115, National Motor Freight Classification.

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